



Moving towards green university: a method of analysis based on multi-criteria decision-making approach to assess sustainability indicators

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Abstract

Increase of greenhouse gases surrounding the earth has recently put the world in a dangerous condition. Thus, the term green building technology has progressively been used to control sustainability matters and to develop standards that can control carbon emissions and energy consumptions. University buildings are not excluded from the energy conservation problems due to their population and activities. Consequently, this study intended to find and prioritize the criteria, sub-criteria, and the associated indicators based on their level of importance in assessing green building universities in Malaysia. Based on the green building index (GBI), the most well-known sustainability rating system for buildings in Malaysia, the criteria and their associated indicators were identified. A panel of experts in consultation and implementation of green buildings in Malaysia was selected for data collection purposes. Criteria and their associated indicators were ranked based on their level of significance using the analytical hierarchy process (AHP). Results showed that indoor environmental quality and energy efficiency were the most significant criteria while the innovation was the least important criteria in evaluating green building universities. Findings of this research provide new knowledge for stakeholders involved in the implementation of green building in universities.

Keywords Sustainability indicators · Green building universities · Malaysia · Indicators for sustainability · Multi-criteria decision making

Introduction

Increase of greenhouse gases (GHG) surrounding the earth has recently put the world in a dangerous condition (Yin and Li 2018). Cutting GHG emissions has been considered as an important concern of governments, specifically for cities with high GHG emissions (Pan and Kao 2021). Thus, both professional and academic parties are looking to find

renewable resources, new technologies, and suitable policies to cut emissions of CO₂ (Wang et al. 2009). It is witnessed that more than one-third of the overall usage of world energy is contributed by buildings and buildings generate more than 40% of total GHG (Shad et al. 2017). In fact, building construction is one of the biggest emitters of GHG as well as one of the major consumers of environmental resources (Liu and Lin 2016). Consequently, diminishing GHG emissions caused by constructions has gradually become the main concern of the building sector (Yin and Li 2018). As results, construction-related establishments now put considerable attention on sustainable/green buildings to maximize the benefits of construction while minimize its harmful effects (Chi et al. 2020). The terms sustainability development and green building have progressively appeared to model the behaviour of buildings and to introduce standards that can control their carbon emission and energy consumption (Mahmood et al. 2021). According to Kibert (2016), green building is “healthy facilities designed and built in a resource-efficient manner, using ecologically based principles”. In addressing the prerequisite of sustainability, the

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term green building is explained as the usage of technologies and methods which are environmentally friendly in the buildings (Love et al. 2011). It is reported that a 30–80% cut in consumption of building energy is manageable by the adoption of green technologies (Yin and Li 2018).

Recently, to encourage green standards in design and construction, different sustainability assessment rating systems have been developed as comprehensive methods for evaluating the level of ‘greenness’ of a building (Shan and Hwang 2018). These rating systems obligated all types of public and private buildings including office, residential, commercial, retail, hospital as well as university to move towards sustainable development rules and regulations (Nilashi et al. 2015; Zou 2019).

Malaysia is considered as the 25th country for GHG emissions around the globe (Mahdiyar et al. 2020). Buildings’ energy consumption showed a sharp increase over the years 1990–2014 and was anticipated to rise in the next 5 years up to 2020 (Suruhanjaya 2014). It is expected that the consumption of energy in Malaysian commercial and residential buildings will be boosted due to the population growth (Asadi et al. 2022; Tabatabaee et al. 2019). Therefore, energy efficiency strategies for buildings in Malaysia are emergency needed to improve the ecological sustainability (Mokhtar Azizi et al. 2013; Shaikh et al. 2017). Malaysia is moving towards a recognized economy, and thus, the nationwide development strategies need to be matched with the world’s megatrends, particularly sustainable initiatives (Ahmed et al. 2021). Therefore, Malaysia has planned to cut 45% of intensity of GHG emission by 2030 in contrast to 2005 status.

Buildings of Malaysian universities are not exempted from the problem of energy conservation due to their population and activities. Construction of the first university buildings in Malaysia dated back to 1960s. However, the majority part of buildings decays and deteriorates due to several issues, including poor workmanship, low-quality materials, over usage as well as insufficient and weak preservation (Ali et al. 2010). According to an official report of Malaysia Ministry of Education (MOE), more than 1 million people are users of university buildings at any specific time (Saleh et al. 2015). High level of energy consumption caused by this population and their activities negatively impact the environment (Mat et al. 2009). Thus, numerous strategies towards sustainable and green university have been planned by MOE to manage and control the energy usage in universities (Foo 2013; Saleh et al. 2015). Meanwhile, universities have been obligated to follow a comprehensive approach to cut the consumption of energy and go green (Choong et al. 2012). Institutions of higher education, as complicated organizations, are considered as model for the society in committing to sustainability practices (Goni et al. 2017). So far, many studies have discovered different aspects of green university (Atici et al. 2021; Fissi et al. 2021; Foo 2013;

Najihah et al. 2013; Saleh et al. 2015); nevertheless, discovering what factors are best suited to assess performance of green building university and ranking their importance level remain largely less explored. Hence, identifying a list of criteria towards green universities building assessment is undeniably required to manage and reduce educational buildings’ energy consumption. List of abbreviations of this study is presented in Table 1.

Research objectives and contributions

Due to the substantial influences that university activities have shown upon society socially and environmentally, there an increasing degree of consideration has been devoted to sustainability matters in higher education by all involved parties around the world (Yuan et al. 2013). Barmeparesos et al. (2019) analysed indoor environmental conditions and energy performance of a student dormitory in Athens. In this study, the building’s energy signature and association between consumption of electricity and the outdoor temperature were assessed. Petidis et al. (2018) conducted a study on the current energy consumption behaviour and indoor conditions of student housing of the Technical University of Crete. They provided five scenarios for dropping consumption of energy toward an approximately zero energy building. Joppolo et al. (2017) study focused on the relationships between building conservation, energy efficiency, and user comfort at the Urbino Colleges. By a suggestion for energy retrofitting, they tried to decrease operational and heating costs.

Literature shows that the initiation of green practices in building universities is still in the initial phases. Current research has mainly focused on the sustainability of university buildings by considering human, energy and environmental aspects. Application of a comprehensive method in the management of university buildings that consider both construction itself and users is significantly lacking (Khalil et al. 2016). Even, enormous rating schemes [such as LEED (Leadership in Energy and Environmental Design), BREEAM (Building Research Establishment’s Environmental Assessment Method), and GBI (Green Buildings Index)] have been proposed and put in practice to simplify the development of green buildings worldwide, the application of such assessment tools to examine the sustainability of university buildings and to prioritize the relative sustainability indicators has been totally overlooked.

Apparently, building sustainability evaluation is a type of MCDM (Multiple-criteria decision making), which is generally founded on knowledge rather than numerical records (Gan et al. 2017; Khoshnava et al. 2018). This decision-making involves various factors to be simultaneously taken into account (Mafarja and Abdullah 2015). AHP (Analytic Hierarchy Process), an MCDM technique, is a well-known method for identifying the rank of factors in imprecision,



Table 1 List of abbreviations

Abbreviation	Description
ACE	Air change effectiveness
AEEP	Advanced EE performance
AHP	Analytic hierarchy process
BEAM	Building environmental assessment method
BRR	Brownfield redevelopment
BUM	Building user manual
CASBEE	Comprehensive assessment system for building environmental efficiency
CDMC	Carbon dioxide monitoring and control
CWM	Construction waste management
DAY	Daylighting
DDCC	Development density and community connectivity
DEMATEL	Decision-making trial and evaluation laboratory
DGC	Daylight glare control
ECAPC	Earthworks—construction activity pollution control
EE	Energy efficiency
EEV	EE verification
ELL	Electric lighting levels
ENC	Enhanced commissioning
ENM	Environment management
EQ	Indoor environment quality
ESM	Electrical sub-metering
ETSC	Environmental tobacco smoke control
EXV	External views
FANP	Fuzzy analytic network process
GBI	Green buildings index
GBIAF	Green building index accredited facilitator
GBRSs	Green building rating systems
GBTool	Green building assessment tool
GHG	Greenhouse gases
GR	Greenery and roof
GVP	Green vehicle priority
HFB	High-frequency ballasts
IAP	Indoor air pollutants
IAQ	IAQ before and during occupancy
IDEDI	Innovation in design and environmental design initiatives
IEB	Industrial existing building
IN	Innovation
INC	Industrial new construction
INL	Internal noise levels
LCA	Life cycle assessment
LIZ	Lighting zoning
MCDM	Multiple-criteria decision making
MEEP	Minimum EE performance
MIAQP	Minimum IAQ performance
MLDS	Metering and leak detection system
MOP	Mould prevention
MR	Materials and resources
MRS	Materials reuse and selection
NREB	Non-residential existing building



Table 1 (continued)

Abbreviation	Description
PAC	Parking capacity
POC	Post occupancy commissioning
POCS	Post occupancy comfort survey: verification
PTA	Public transportation access
QLASSIC	Quality assessment system for building construction work
RAH	Rainwater harvesting
RCA	Refrigerants and clean agents
RCM	Recycled content materials
REE	Renewable energy
REM	Regional materials
RNC	Residential new construction
SCR	Storage and collection of recyclables
SDQ	Stormwater design—quantity and quality control
SEM	Structural equation modelling
SM	Sustainable site planning and management
SUM	Sustainable maintenance
SUT	Sustainable timber
TCDCS	Thermal comfort: design and controllability of systems
WAR	Water recycling
WE	Water efficiency
WEF	Water-efficient fittings
LEED	Leadership in energy and environmental design
WEIL	Water-efficient—irrigation/landscaping
NRNC	Non-residential new construction
WSA	Workers' site amenities

vagueness, and uncertain settings, particularly in environmental-related fields (Cegan et al. 2017; Leong et al. 2020; Shad et al. 2017). This method is based on powerful mathematics which increases objectivity and transparency through pairwise comparisons (Khalil et al. 2016). Since the limited number of efforts has focused on evaluating green building universities using MCDM techniques, this research aims to take the advantages of AHP to assess the green building universities in Malaysia. Consequently, the purposes of this study are:

1. To discover the criteria that are vital to assess the green building universities.
2. To rank the criteria and the associated items based on their level of significance in evaluating the universities' green building using AHP technique.

This research provides the following contributions:

1. The same as many corporate entities and businesses, higher education organizations are progressively motivated by their external and internal partners to represent their involvement in having a sustainable society

(Tans 2017). Therefore, different levels of authorities are extensively involved in providing long-term strategic plans in order to attain sustainability through performance optimization, and thus, application of different green building evaluation methods has been promoted (Khalil et al. 2016; Ragazzi and Ghidini 2017). This study can be considered as an initial reference for different parties of university green building development.

2. So far, the modern nations have attained reasonable progress in managing environment efficiency by utilizing well-organized assessment tools and sustainable practices (Asadi et al. 2021). However, developing countries are still in the first steps and expanding sustainability practices is significant for them (Ali and Al Nsairat 2009). Consequently, this study can offer broader insights and understandings of green building university in a developing country such as Malaysia.
3. This research assesses and prioritizes a comprehensive set of criteria, sub-criteria and indicators for the first time in the context of green universities in Malaysia using expert's points of view. This offers a clear understanding to all stakeholders on the significant criteria that require proper attention.



4. Currently, only a limited number of corporations are providing sustainability consultations and services for the construction sector in Malaysia. However, Malaysia has shown potential to greatly profit from this market growth (Papargyropoulou et al. 2012). It is expected this research will consider as a baseline for the development of sustainability market in Malaysian built environment sector.
5. So far, AHP has been widely used to rank the criteria and sub-criteria based on their level of importance in vagueness, imprecision, and uncertainty settings in different fields (Calabrese et al. 2019; Nilashi et al. 2015). This research is among the first efforts that employed the Fuzzy AHP technique to evaluate green buildings in university context.

The structure of the current research is as follow. Literature review is provided in Sect. 2. The research method is described in Sect. 3. Section 4 represents data analysis and results. Discussions are represented in Sect. 5. Finally, conclusions are provided in Sect. 6.

Literature review

In this section, the most famous green building rating schemes are introduced and discussed. Next, Malaysian GBI rating system is presented. Finally, a summary of the research on green building evaluation using MCDM methods is provided.

An overview of GBRs (green building rating systems)

Conventional buildings are constructed and used according to specific criteria and codes. Green buildings make headway by considering sustainability goals in carbon emission decline, energy-saving, and indoor air quality enhancement. As a result, GBRs have been introduced to assess and verify green buildings (Illankoon and Lu 2019). LCA (Life Cycle Assessment) and criteria-based rating are two widely-recognized assessment methodologies used in GBRs for measuring the level of greenness of buildings (Nilashi et al. 2015). LCA evaluation systems have normally been utilized to select appropriate building design and material as well as local facility opportunities such as energy supply, transport type, and management of waste in the process of design (Malmqvist et al. 2005). In contrast, criteria-based rating systems are regional and national GBRs that are used for evaluating eco-buildings in different counties considering the differences in culture, geographical characteristics, potential energy resources, and climatic conditions (Shad et al. 2017). BREEAM,

LEED, CASBEE (Comprehensive Assessment System for Building Environmental Efficiency), and GBTool (Green Building Assessment Tool) of Multinational Cooperation are the available prominent GBRs. These assessment systems are based on similar classifications. However, each tool emphasizes on the certain assessment criteria considering the situation of the country (Nilashi et al. 2015; Shad et al. 2017).

In 1990, the building research establishment in the UK launched BREEAM. It was initially developed to focus on environmental characteristics, however, later it has emphasized on social and economic aspects as well (Haroglu 2013; Serrano-Baena et al. 2020). The BREEAM rating method encompasses all substantial aspects from creation and preparation to in-use during the whole lifecycle of a building. BREEAM takes into consideration the wellbeing and health, management, water, energy, transport, land use and ecology, materials, waste, innovation, and pollution categories to assess sustainable value. The final score of a building is computed a total of each of the categories considering their assigned rates. Based on the final score and rating scale, the type of certification is assigned (Mattoni et al. 2018).

The LEED has shown the best market penetration worldwide. So far, 165 countries including China, US, Brazil, and India have subscribed to this system (Chi et al. 2020; MacNaughton et al. 2018). LEED segments its measures into certain areas: “Location and Transportation”, “Sustainable Sites”, “Water Efficiency”, “Energy and Atmosphere”, “Materials and Resources”, “Indoor Environmental Quality”, “Innovation” and “Regional Priority”, and “Innovation” (Champagne and Aktas 2016). Buildings are regularly evaluated across all aspect’s integral to their environmental performance. By reviewing the technical process and checking the overall credits assigned to a project, the certification level is awarded to a building.

GBTool was developed in 1996 in Canada. This system is based on a common toolbox that can be modified based on the needs and requirements of local building performance. This method employs a scoring technique ranging from “Demanding Performance” to “Unsatisfactory Performance” (Rosa and Haddad 2013; Shad et al. 2017).

In 2001, Sustainable Building Consortium founded CASBEE in Japan. The main objective of developing this system was to establish eco-friendly constructions by considering environmental performance within the building lifetime. This system calculates building scores from a formulation that is resulted from the form that the users complete for their own building. Then buildings are assigned to one of the predefined classes of greenness. CASBEE has represented outstanding performance in constructing sustainable buildings in Japan (Shad et al. 2017). Table 2 highlights the main characteristics of BREEAM, LEED, GBTool, CASBEE rating systems.



Table 2 Characteristics of popular GBRs (Shad et al., 2017)

Rating system	Country	Year	Evaluation procedure	Core criteria
BREEAM	UK	1990	Mixture of feature-specific and performance-based criteria	Management wellbeing and health, energy, materials, transport, ecology and land use, pollution, waste, innovation, water
LEED	USA	1998	Energy cost budget approach and feature-specific criteria	Atmosphere and energy, indoor environmental quality, water efficiency, transportation and location, sustainable sites, regional priority, innovation, resources, and materials
GBTool	Canada	1996	–	Project development and planning, Site selection, indoor environmental quality, consumption energy and resource, environmental loadings, service quality, perceptual and cultural attributes, economic and social and attributes
CASBEE	Japan	2001	Extremely complicated weighting system utilized at every level	Efficiency of resource, energy efficiency, local atmosphere, indoor atmosphere

Green building index (GBI) in the context of Malaysia

In 2009, the Malaysia's GBI (Green Building Index) was launched at Convention Centre Kuala Lumpur. The GBI was created upon the current rating systems and has been greatly revised to be matched with the tropical climate, social and cultural preferences, and environmental context of Malaysia (Yusoff and Wen 2014). GBI has become the primary wide-ranging rating schema to assess the buildings' performance in Malaysia. It is aimed to foster sustainability in the building sector and improve consciousness about environmental issues among all parties including architects, developers, designers, engineers, planners, and the Public. This assessment system can assist stakeholders to design and construct green buildings by considering improved accessibility to public transport, water and energy savings, improved indoor environment, and the utilization of greenery and reusing for their developments. GBI offers a shared standard and language of assessment, to modify the built environment to decrease its harms for the environmental, and to warrant present buildings are upgraded and renovated appropriately to meet criteria in the future (Mun 2009).

Buildings are granted the GBI Malaysia rating according to six core criteria EE (Energy Efficiency), EQ (Indoor Environment Quality), SM (Sustainable Site Planning and Management), MR (Materials and Resources), WE (Water Efficiency), and IN (Innovation) with total points of 100.

EE: Enhance consumption of energy by improving building location, minimizing solar heat achieve on the building facades, storing natural lighting, utilizing the most appropriate strategies in buildings such as usage of renewable energy and ensuring the frequent maintenance.

EQ: Attain excellent quality performance in acoustics, visual and thermal comfort, and indoor air quality.

SM: Picking suitable locations with access to community services, public transportation, open spaces, and landscaping. Preventing areas, which are environmentally sensitive across the revitalization of existing sites and brownfields.

MR: Support usage of materials that are environmental friendly.

WE: Harvesting rainwater, reprocessing water, and fittings water-saving.

IN: Advanced design which considers the aims and scope of the GBI.

Collecting points in the above dimensions implies that the building is expected to be more sustainable in comparison to those which do not consider the concerns. According to this rating system, 86–100 points, 76–85 points, 66–75 points, and 50–65 points are considered as Platinum, Gold, Silver, and Certified building classification, respectively (Green-buildingindex 2009a).

Related work on green building evaluation using MCDM

MCDM is generally involved in revealing the level of importance of determinants with the aim of solving complicated multiple decision-making problems (Safari et al. 2015). From this viewpoint, every assessment of sustainability is in fact an MCDM considering social, environmental, and economic matters (Gan et al. 2017; Munda 2005; Shad et al. 2017). According to Cegan et al. (2017), the application of MCDM in environmental science research is significantly growing. So far, MCDM methods have been successfully utilized in many studies to assess sustainability and to provide a direction towards the success sustainable future (Gandhi et al. 2018; Yazdani et al. 2017). Recently, Yang and Zhang (2021) proposed an MCDM framework to evaluate sustainable urban drainage systems. Considering multi-criteria nature of the performance assessment system, this study



utilized MCDM analysis to stabilize the objectivity and completeness. Meanwhile, a large number of research have been conducted to measure the sustainability and greenness of buildings using different courses of MCDM techniques (Table 3).

Utilizing fuzzy DEMATEL (Decision-Making Trial and Evaluation Laboratory), Yadegaridehkordi et al. (2020) examined sustainability criteria for evaluating sustainable buildings in manufacturing. This study collected data from a panel of professionals. Findings revealed that EE and EQ were the most significant factors in the green building assessment of Malaysian manufacturing.

Khoshnava et al. (2018) assessed green building material-related indicators by employing a hybrid MCDM method. By considering two steps DEMATEL and FANP (Fuzzy Analytic Network Process), they focused on achieving health-promoting, ecological, and high-performance materials for building in the construction industry in Malaysia. Yin and Li (2018) proposed a new model for matching technologies related to the green building. Considering intuitionistic fuzzy sets and utilizing associations between factors, they generated a two-sided matching model and results which are steadier with the reality.

Ulubeyli and Kazanci (2018) developed a sustainability assessment model for green building business in Turkey. To verify the assessment method, fuzzy synthetic evaluation was used and green building industry's macro-environmental conditions were measured. Shad et al. (2017) designed a green building evaluation system in Iranian context. They employed AHP approach to rank all criteria and their associated sub-criterion by calculating the weights of criteria. The developed tool was tested on case study offices in one of the biggest metropolis in Iran, Mashhad. Tupenaite et al. (2017) proposed an assessment system for the sustainability of new housing developments in the Baltic States. They also performed fuzzy AHP to determine the significance of determinants in green buildings. Mardani et al. (2016) designed a framework with the aim of assessing major techniques for

saving energy in hotels of Iran. Following both quantitative and qualitative methods, they used fuzzy Delphi and AHP to rank performance according to the likeness to perfect solution. Finally, they reported that system and equipment effectiveness and decrease of cooling and heating demands were the most important groups for saving energy. Nilashi et al. (2015) developed a new method to measure the level of a green building performance based on knowledge-based expert system. They utilized fuzzy logic and AHP in proposing the expert system. Based on the assessment categories, the high-ranked parameters were nominated to be used as input in the fuzzy logic model and to evaluate the green buildings' level of performance from SE2 perceptions.

A weighting and rating tool for Chinese green store buildings was developed by Yu et al. (2015). They tried to avoid the impact of "imbalanced performance" in the process of green buildings labelling. They used AHP for the purpose of ranking dimensions. Finally, they ranked operation management, indoor environmental quality, and energy efficiency as the main involving criteria. A study was conducted by Chandratilake and Dias (2013) among building construction segments in order to design a buildings sustainability rating system in Sri Lanka. They built a system based on specific rating domains. They employed AHP and direct ranking techniques for ranking the indicators of each dimension. Next, the total weights and weights calculated by other rating tools were compared. Alyami et al. (2013) claimed that the popular international rating tools are not fit the context of Saudi Arabia. Thus, they aimed to propose a new assessment system for the built environment in this country. They categorized the assessment indicators into nine groups. Next, Delphi technique was performed by involving experts from government, academia, and industry to list the sustainable residential building criteria.

So far, a variety of sustainability tools and systems have been utilized for the evaluation of buildings' sustainability in different notions. However, sustainability indicators for green building university in Malaysia are still not well explored. According to Chandratilake and Dias (2013),

Table 3 Existing studies on green building evaluation

References	Evaluation technique	Building type	Country
Yadegaridehkordi et al. (2020)	Fuzzy DEMATEL	Manufacturing	Malaysia
Khoshnava et al. (2018)	DEMATEL + FANP	Industry	Malaysia
Yin and Li (2018)	Intuitionistic fuzzy sets	Enterprises	China
Ulubeyli and Kazanci (2018)	fuzzy synthetic evaluation	Business	Turkey
Shad et al. (2017)	AHP	Office	Iran
Tupenaite et al. (2017)	fuzzy AHP	Residential	Baltic States
Mardani et al. (2016)	fuzzy Delphi + AHP	Hotel	Iran
Yu et al. (2015)	Expert Group Decision AHP	Store	China
Nilashi et al. (2015)	fuzzy AHP	Hospital	Malaysia
Chandratilake and Dias (2013)	AHP	–	Sri Lanka
Alyami et al. (2013)	Delphi technique	Residential	Saudi Arabia



sustainability indicators and their level of importance vary based on environmental conditions as well as social and economic issues of the countries.

It is determined that the AHP is widely utilized and the most appropriate method in green building evaluation studies. Meanwhile, AHP has been extensively and successfully applied to assess complex problems in other fields (Baharuddin et al. 2016; Li et al. 2020; Yadegaridehkordi et al. 2018). AHP is a commonly employed MCDM method to identify the weights of criteria and rank the alternatives in a regulated approach using pairwise comparison. It surpasses by constructing problems analytically, ease of use, and assessing both weights of criteria and alternative priorities (Liu et al. 2020). This method can convert the complicated decision-making process into an easy to solve the problem (Wang et al. 2020). In addition, fuzzy AHP performs well in the situations that the importance level of the multiple criteria is close to each other (Li et al. 2020).

Hence, considering GBI, the primary wide-ranging rating system in Malaysia, this research intended to employ the Fuzzy AHP technique to evaluate green building university in Malaysia.

Materials and methods

In the current research, AHP, a well-known MCDM method, was employed to identify the rank of factors in each group. AHP has been a successful technique in the assessment of green buildings based on GBRSSs (AbdelAzim et al. 2017; Yu et al. 2015). AHP has shown enough power in managing complex and ambiguous decisions while following a simple assessment procedure.

AHP survey questionnaire was designed for collecting data in this study. Based on the identified criteria and rules suggested by Saaty (1990), questions were developed on 9-point scale to be used for pairwise comparisons. The use of 9-point scale is effective in assessing the main criteria and sub-criteria as it could be better used by the experts for their verbal judgements. A panel of experts including academics and professionals, who had experience and knowledge in consultation and implementation of green buildings in Malaysia, has been selected. It is noted that the email and contact information of the respondents were found from Google and the official website of the universities. The online questionnaire survey was based on two main parts, experts' demographic information and the questions based on the key criteria and their associated sub-criteria. Link of the online survey along with the explanations of the aim and scope of the study emailed the respondents. Reminder emails were sent to the respondents every 2 weeks. Finally, after 2 months, a total of 20 professionals returned the complete questionnaires. The demographic information of

Table 4 Experts' characteristics

Category	Characteristic	Number	Percentage (%)
Gender	Male	15	75
	Female	5	25
Position	Professor	9	45
	Associate professor	8	40
	Consultant	3	15
Experience	< 10 years	3	15
	10–14 years	6	30
	15–20 years	7	35
	< 21 years	4	20

Table 5 Scale comparison of AHP

Weight	Definition
1	"Equal importance"
3	"Moderate importance"
5	"Strong importance"
7	"Very strong importance"
9	"Absolute (extreme) importance"
2, 4, 6, 8	"Intermediate values"

respondents is presented in Table 4. According to this table, the majority of the participants were male (75%). 35% of the experts had 15–20 years of related working experience.

AHP

In this study, 1–9 scale measures were used to collect the data (see Table 5). The AHP procedure is shown in Fig. 1.

In AHP, the pairwise comparisons are presented in square matrices. As an example, in the following square matrix $R_{N \times N}$ which includes N criteria, the respondents provide the pairwise comparisons for criteria C_i in relation to the criteria C_j . The responses a_{ij} indicate the comparative importance of criteria C_i with respect to attribute C_j . In $R_{N \times N}$, $a_{ij} = 1$ once $i = j$ and for all entries in the matrix $R_{N \times N}$, the $a_{ji} = 1/a_{ij}$.

$$\begin{array}{c} \text{Criteria} \end{array} \quad \begin{array}{cccccc} C_1 & C_2 & C_3 & \cdots & \cdots & C_N \end{array} \\
 \begin{array}{c} C_1 \\ C_2 \\ C_3 \\ \vdots \\ C_N \end{array} \quad \begin{bmatrix} 1 & a_{12} & a_{13} & \cdots & \cdots & a_{1N} \\ a_{21} & 1 & a_{23} & \cdots & \cdots & a_{2N} \\ a_{31} & a_{32} & 1 & \cdots & \cdots & a_{3N} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{N1} & a_{N2} & a_{N3} & \cdots & \cdots & 1 \end{bmatrix} \quad (1)$$

For each criterion, the normalized weight (w_j) is found using the geometric mean of the i th row in the matrix and then by geometric means normalization of rows in the matrix $R_{N \times N}$. This is done using Eqs. (2) and (3).

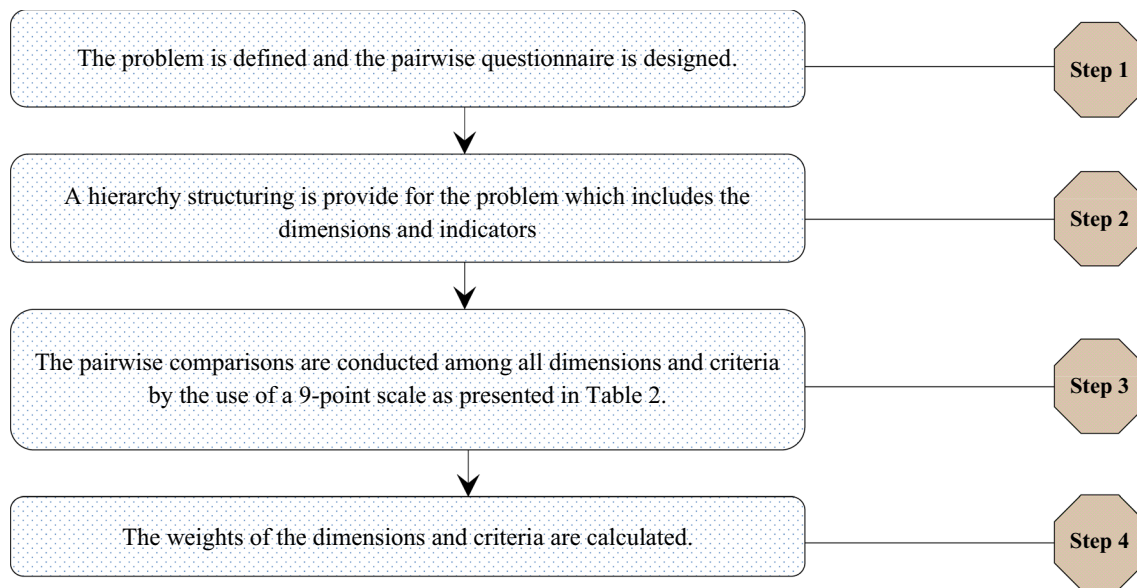


Fig. 1 AHP procedure

$$G_j = \left[\prod_{i=1}^N a_{ij} \right]^{1/N} \quad (2)$$

$$w_j = \frac{G_j}{\sum_{j=1}^N G_j} \quad (3)$$

In AHP, the consistency for a pairwise comparison matrix is calculated by consistency ratio. The consistency ratio for all pairwise comparison matrices must be less than 10%.

Results and discussion

Expert Choice (<http://www.expertchoice.com/>) software was used to employ the AHP technique on the collected data. In Malaysia, there are fourteen GBI tools, which are all founded on six core criteria (EE, IN, WE, MR, SM, EQ), available for different types of constructions (see Fig. 2).

In this study, NRNC tool was employed to evaluate the green buildings' sustainability in universities in Malaysia. Each of the core criteria is split in the subsequent sub-criteria and indicators to accumulate the needed credit points. EE is comprised of design, commissioning, verification, and maintenance sub-criteria and nine items with maximum points of 35. EQ is comprised of air quality, lighting, thermal comfort, verification, and visual and acoustic comfort sub-criteria and 15 items with maximum points of 21. SM is comprised of transportation, site planning, construction management, and design sub-criteria and 13 items with maximum points of 16. MR is comprised of sustainable

resources, reused and recycled materials, waste management, and green products sub-criteria and seven items with maximum points of 11. WE is comprised of water harvesting & recycling and increased efficiency sub-criteria and five items with maximum points of 10. IN with two items and maximum points of 7 (refer to the Appendix 1 for details of criteria, items' definitions, and points). Table 6 shows the core green building index assessment criteria, sub-criteria, indicators, and max points for NRNC.

The hierarchy structure of the assessment via AHP consisted of 6 core groups, 17 sub-groups, and 51 indicators, according to Table 6. Therefore, pairwise questions were designed to explore the importance level of an item in comparison to the others in each group and its sub-groups by performing pairwise comparisons. It is obvious that the number of criteria formulates the number of pairwise comparisons. The matrices of pairwise comparison for this study are presented in Appendix 2. These matrices are calculated based on 1–9 scale measures for main groups and sub-groups by linking to the linguistic phrases in Table 4. From the results of comparisons in Appendix 2, it can be seen that all comparisons completed by the experts are valid by a Consistency Ratio less than 0.1. Table 7 shows the final weights for all core criteria, sub-criteria and indications.

According to the findings, EE (weight = 0.342) and EQ (weight = 0.206) are the most vital dimensions in GBI-NRNC based on the experts' opinions. "Design" (weight = 0.60) is the most critical sub-criteria in EE dimension and LIZ (weight = 0.421) is the most vital indicator in "Design". "Lighting, Visual & Acoustic Comfort" (weight = 0.461) is the most critical factor in EQ dimension. Both DAY and EXV (weight = 0.250) show



Fig. 2 GBI assessment systems are based on six main factors (Yadegaridehkordi et al., 2020).

Note: NREB non-residential existing building, *NRNC* non-residential new construction, *INC* industrial new construction, *IEB* industrial existing building, *RNC* residential new construction

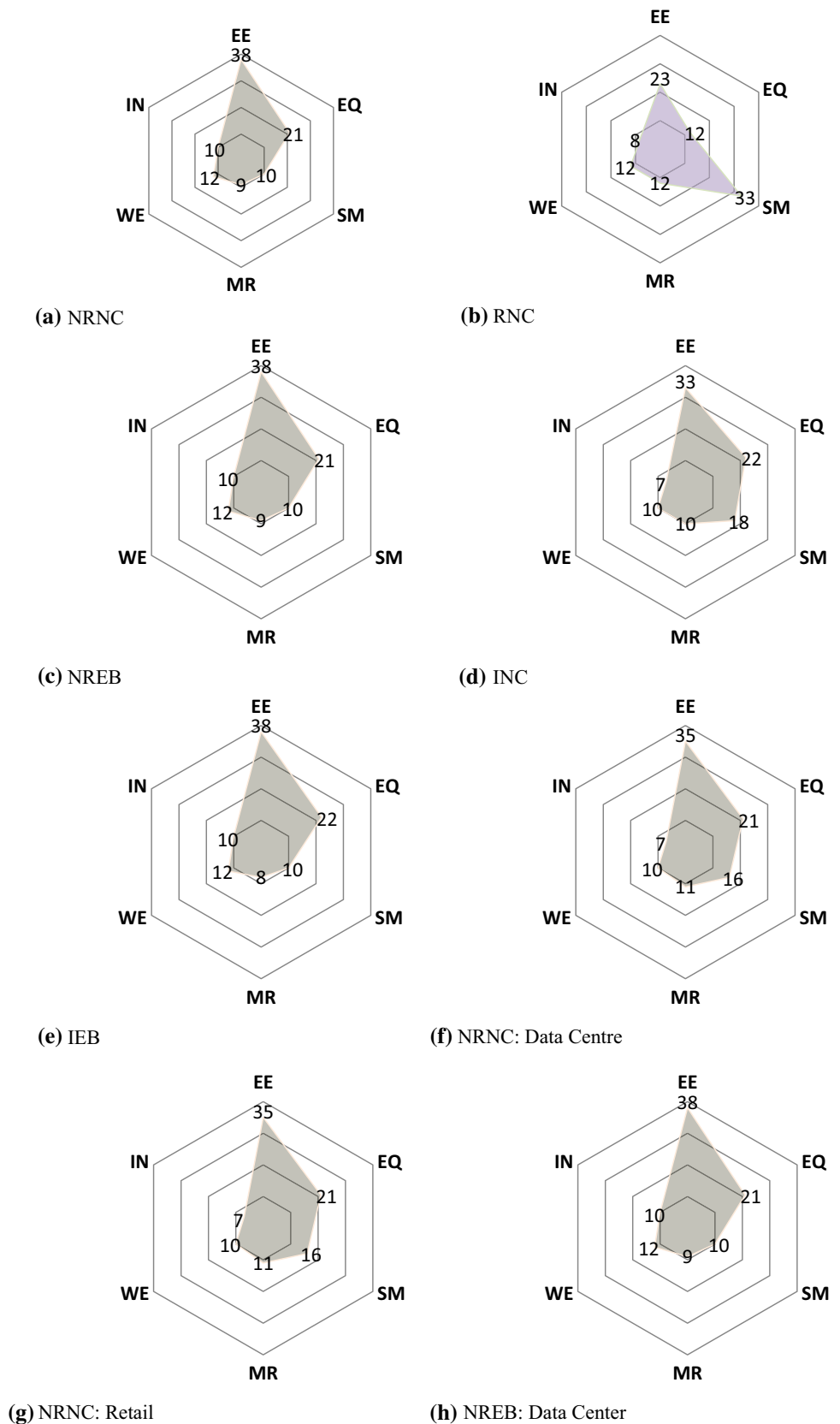
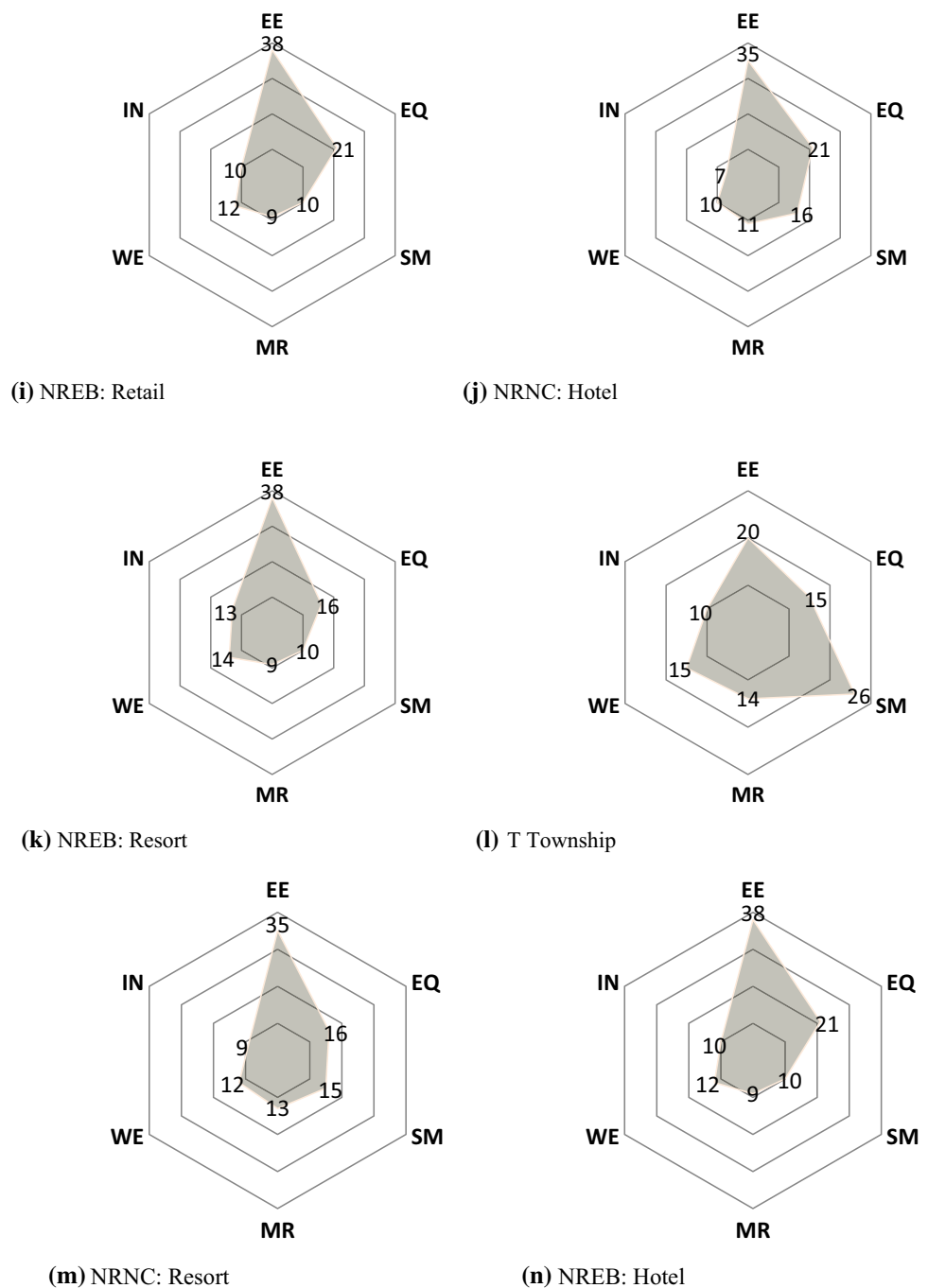


Fig. 2 (continued)



the largest importance in this sub-group. “Site Planning” (weight = 0.395) is the most critical factor in SM dimension. SIS and BRR (weight = 0.333) are realized as two significant items of this sub-criteria. “Reused and Recycled Materials” (weight = 0.444) is the first ranked factor in MR dimension. RCM and MRS indicators (weight = 0.50) show equal importance in this sub-group. “Increased Efficiency” (weight = 0.80) is the most vital factor in WE dimension. WEIL, WEF and MLDS (weight of 0.333) show similar importance in this sub-group. Finally, IN is the least

important factor based on experts’ opinion. IDEDI (weight 0.857) gains the highest importance level in IN dimension.

Discussion

Results of this research indicated that EE was the most substantial sustainability dimension in assessing universities’ green building. In this category “Design” was found to be the most important sub-criteria and LIZ was the most



Table 6 Assessment criteria in NRNC

No	Criteria	Sub-criteria	Item (indicator)
1	EQ	Air quality	MIAQP
			IAP
			MOP
			CDMC
			ETSC
		Thermal comfort	ACE
			TCDCS
		Lighting, visual and acoustic comfort	DAY
			HFB
			ELL
			DGC
			INL
			EXV
			IAQ
			POCS
2	EE	Design	ESM
			LIZ
			REE
			AEEP
			MEEP
		Commissioning	ENC
			POC
		Verification and maintenance	EEV
			SUM
			BRR
3	SM	Site planning	SIS
			ENM
			DDCC
			QLASSIC
		Construction management	WSA
			ECAPC
			GVP
		Transportation	PAC
			PTA
			SDQ
		Design	BUM
			GR
			RCM
4	MR	Reused and recycled materials	MRS
			REM
		Sustainable resources	RCA
			SCR
		Waste management	SUT
			CWM
			WAR
5	WE	Water harvesting and recycling	RAH
			WEF
		Increased efficiency	WEIL
			MLDS

Table 6 (continued)

No	Criteria	Sub-criteria	Item (indicator)
6	IN		IDEDI
			GBIAF

ACE air change effectiveness, *AEEP* advanced EE performance, *BRR* brownfield redevelopment, *BUM* building user manual, *CDMC* carbon dioxide monitoring and control, *CWM* construction waste management, *DAY* Daylighting, *DDCC* development density and community connectivity, *DGC* daylight glare control, *ECAPC* earth-works—construction activity pollution control, *EEV* EE verification, *ELL* electric lighting levels, *ENC* enhanced commissioning, *ENM* environment management, *ESM* electrical sub-metering, *ETSC* environmental tobacco smoke control, *EXV* external views, *GBIAF* green building index accredited facilitator, *GR* greenery and roof, *GVP* green vehicle priority, *HFB* high frequency ballasts, *IAP* indoor air pollutants, *IAQ* indoor air pollutants, *IDEDI* innovation in design and environmental design initiatives, *INL* internal noise levels, *LIZ* lighting zoning, *MEEP* minimum EE performance, *MIAQP* minimum IAQ performance, *MLDS* metering and leak detection system, *MOP* mould prevention, *MRS* materials reuse and selection, *PAC* parking capacity, *POC* post occupancy commissioning, *POCS* post occupancy comfort survey: verification, *PTA* public transportation access, *QLASSIC* quality assessment system for building construction work, *RAH* rainwater harvesting, *RCA* refrigerants and clean agents, *RCM* recycled content materials, *REE* renewable energy, *REM* regional materials, *SCR* storage and collection of recyclables, *SDQ* stormwater design—quantity and quality control, *SUM* sustainable maintenance, *SUT* sustainable timber, *TCDCS* thermal comfort: design and controllability of systems *WAR* water recycling, *WEF* water efficient fittings, *WEIL* water efficient—irrigation/landscaping, *WSA* workers' site amenities

important indicator. Similar to this finding, Illankoon et al. (2017) believed that 'Energy' factor has received the maximum attention in all world-famous green building rating tools (LEED, BREEAM, etc.). This finding is also in line with Qian and Chan (2010) who considered energy efficiency improvement as one of the main important factors that can diminish GHG emissions of the entire construction life cycle. Meanwhile, Yu et al. (2015) ranked EE as second significant factor within green store buildings in China. Abd Rahman et al. (2021) pointed out that EE is an important part of all green building rating schemas.

EQ was ranked as the second important dimension with "Lighting, Visual and Acoustic Comfort" as its vital subgroup and DAY and EXV high-ranked indicators. This finding is in accordance with Yu et al. (2015) findings that ranked indoor environmental quality as the first influencer of stores' green building. However, since the scope of this study was store buildings, indoor lighting environment, indoor acoustic environment, indoor thermal environment, and indoor air quality were considered as sub-criteria of the indoor environment. It is found that indoor comfort and environmental quality are the most essential factors influencing green buildings in Turkey. Opposite to this finding, Khalil et al. (2016) confirmed that indoor environmental



Table 7 Calculated weights by AHP

No	Criteria	Sub-criteria	Item (indicator)	Weights			
1	EE (0.342)	Design (0.60)	ESM	0.092			
			LIZ	0.421			
			MEEP	0.170			
			AEEP	0.092			
			REE	0.226			
		Commissioning (0.20)	POC	0.25			
			ENC	0.750			
		Verification and maintenance (0.20)	SUM	0.80			
			EEV	0.20			
2	EQ (0.206)	Air quality (0.236)	MOP	0.167			
			ETSC	0.167			
			MIAQP	0.167			
			MOP	0.167			
			IAP	0.333			
		Thermal comfort (0.135)	TCDCS	0.750			
			ACE	0.250			
			Lighting, visual and acoustic comfort (0.461)	ELL	0.125		
		DGC		0.125			
		DAY		0.250			
		EXV		0.250			
		HFB		0.125			
		INL		0.125			
		POCS		0.50			
		Verification (0.168)	IAQ	0.50			
		3	SM (0.180)	Site planning (0.395)	BRR	0.333	
					SIS	0.333	
					ENM	0.167	
DDCC	0.167						
QLASSIC	0.333						
Construction management (0.163)	ECAPC			0.333			
	WSA			0.333			
	Transportation (0.163)			GVP	0.333		
PAC				0.333			
PTA				0.333			
Design (0.278)				GR	0.50		
	BUM			0.25			
	SCR			0.25			
	4			MR (0.117)	Sustainable resources (0.122)	SUT	0.50
						REM	0.50
Waste management (0.312)					CWM	0.667	
					SCR	0.333	
Reused and recycled materials (0.444)					RCM	0.50	
	MRS			0.50			
	Green products (0.122)	RCA	1				
5	WE (0.101)	Water harvesting and recycling (0.20)	WAR	0.50			
			RAH	0.50			
		Increased efficiency (0.80)	WEF	0.333			
			MLDS	0.333			
			WEIL	0.333			
6	IN (0.053)	–	IDEDI	0.857			
			GBIAF	0.143			



performance has less rank in compared to functional and technical performance in the situation of users' risk and building performance. Shad et al. (2017) ranked EQ as the least important assessment dimension in Iranian offices' green buildings.

This study ranked WE and IN as less important sustainability indicators. This result is in accordance with Khoshnava et al. (2018) that found water conservation is the last green building material criteria that need to be considered. Likewise, Yu et al. (2015) reported that in comparison to other criteria water efficiency has less priority in assessing green store buildings in China. According to Shad et al. (2017), innovation has the lowest score while water efficiency stands among the highest-ranked factors in assessing green office buildings in Mashhad. They also found that most nations do not consider innovation item as a vital factor in the assessment procedure of green building (Shad et al. 2017).

Referring to Yu et al. (2015), green building assessment tools normally insist on different criteria by considering buildings' characteristics such as location and indoor environmental quality necessities, function, layout, building size, operation management, and resource and energy efficiency. Normally for residential buildings emphasis is on outdoor landscape and indoor comfort; for manufacturing buildings, the emphasis is on industrial pollution and manufacturing costs; and for workplace buildings emphasis is on recycling of source as well as energy efficiency. Due to specific characteristics of university buildings in Malaysia, such as high occupant density by students and staffs and intensive use of air conditioning almost all year long, in this study higher priority has been given to EE and EQ criteria.

Conclusion

Since the twentieth century, in a response to efforts for energy efficiency practices, many green building evaluation systems have been developed (Papargyropoulou et al. 2012). Following this trend, Malaysian government aims to significantly decrease the GHG emissions intensity by the year 2030 by releasing numerous related rules and regulations (Hamid et al. 2014; Lim et al. 2017). Consequently, all sectors including universities have been obligated to follow a full approach to cut the consumption of energy and go green (Choong et al. 2012). In this way, GBI is the most applied sustainability rating tool designed to address building sustainability in the Malaysian building industry. However, despite the popularity and significant role of GBI in getting involved in green initiatives, research concerning the investigation of the GBI criteria and their associated sub-criteria in university building performance remain greatly limited. More specifically, ranking the level of importance

of criteria and their associations is totally overlooked in the literature. Therefore, considering GBI, this research intended to discover and rank the criteria and associated indicators to evaluate the universities' green building in Malaysia using experts' opinion and AHP approach. According to the findings, EQ and EE are the most vital dimensions in comparison to SM, MR, WE, and IN. Although all the criteria are critical, prioritizing them would assist government, stakeholders, and policymakers, to understand which criteria are worth focusing more consideration on. Thus, they can develop appropriate administrative guidelines by focusing on the most critical factors for moving towards green program in shorter-term.

It is reported that 14.3% of the total used energy is in Malaysia is by buildings and 53% of electrical energy is being used in commercial and residential subdivisions (Shaikh et al. 2017). Therefore, EE in buildings is vital in to control the consumption of energy and expand the overall environmental greenness. However, the essential concept is to least cost of energy and resources while providing a relaxed and healthy living condition (Yu et al. 2015). In this regard, government should regulate the prices of energy to fulfil the real requirements of the construction industry. They may need to consider different levels of energy pricing for different universities. Appropriate building energy efficiency standards need to be utilized through legislation to clarify the authorized responsibilities of different stakeholders involved. Economic incentives can be provided by the government to inspire the building energy efficiency market. Meanwhile, financial supports should be allocated to target building energy efficiency enhancements by encouraging fundamental development projects as well as technological innovations. Building energy efficient related education courses/seminars should be organized for both public and building professionals to widely create demand for and gratefulness of building energy-efficiency products. On the other hand, usage of green building materials with lower or no negative influence on the environment and humans can significantly decrease building energy consumption and CO₂ emission (Khoshnava et al. 2018). Thus, the choice of building materials should be based on standard sustainability criteria which subsequently guarantee a sustainable development in building design and construction, for example, by insisting on energy-efficient materials produced with renewable or natural resources. Meanwhile, EE in buildings can be meaningfully controlled by incorporating appropriate green building technologies such as solar system, green roof and wall technology, and heat pump. Meanwhile, there is an immediate demand for accurate prediction and monitoring of buildings' energy consumption utilizing the modern technologies including data analytics and Internet of Things (IoT) to improve energy usage (Goudarzi et al. 2021). Application of such technologies not only proposes economic, social,



and environmental sustainability benefits but also enhances EQ, enhances productivity, and increases WE in buildings (Yin and Li 2018). Therefore, great consideration should be given to expand adoption of green building technologies in building sector.

According to the literature, EQ has significant impact on residents' wellbeing, comfort, and output in workplace environments (Al Horr et al., 2016a, 2016b). Thus, it is vital to emphasize on the EQ of the official environments by providing suitable and comfortable places for the staffs. Green building design strategies and techniques can expressively enhance the indoor office condition by offering an atmosphere which matches residents' performance and comfort in both new and reconstructed buildings (Singh et al. 2010). Since artificial lighting is the main producer of carbon dioxide emissions (Busch et al. 1993), intelligent lighting systems and effective daylighting designs should be utilized in building to reduce the load of artificial energy thus reduce the carbon footprint of a building. To do so, smart daylighting controls such as continuous dimming and switching control need to be provided in buildings. Meanwhile, by considering indoor and outdoor light levels, sky luminance, and solar position and radiation, enough number of windows should be embedded in the buildings in order to make benefits of more natural light/sunlight. It is recommended to government and green policymakers to organize special programs with the aim of improving knowledge related to the characteristics of the indoor environment in order convince all involved parties that decision to renovate or modernize a building is a useful investment, highlighting that the improvement of indoor environment quality implies a growth of occupants' productivity.

Generally, this study provides a nutshell of how GBI should be implemented in practice and shares new insights and knowledge on greening education pillars in Malaysia and the other developing countries. In fact, considering controlled legislation in order to make sure all new constructions follow standard green certifications such as GBI is compulsory requirement that can move the Malaysian industry towards sustainable design and surge the number of green buildings delivered. As the Malaysian sustainability market is still in developing stage, more qualified specialists are needed to enter to this market (Papargyropoulou et al. 2012). Thus, more training programs should be delivered by relevant experts' bodies and higher education institutions by special focus on GBI and its related dimensions and indicators in order to train highly qualified future specialists.

Limitations and future work

This research evaluated the level of importance of sustainability criteria for universities' green building, however, still more qualitative and quantitative investigations with emphasis on single criteria and/or indicator are required to critically analysis and promote development of green universities practice in Malaysia. This study is limited to ranking of criteria while overlooking the interdependence of them. This study used an online questionnaire survey and performed MCDM method to identify and prioritize the criteria, sub-criteria, and the associated indicators based on their level of importance in assessing green building university. Following Yang and Zhang (2021), an integration of MCDM method and mathematical quantification by the use of interviews is recommended for future studies in order to balance objectivity and comprehensiveness in the assessment. In addition, this study did not include any alternative to be assessed by the AHP technique. Therefore, as sensitivity analysis (Chen et al. 2021; Song et al. 2021) which is a critical stage in developing models with the MCDM techniques in determining whether the solution is implementable and robust, was not investigated in this research. This should be considered in the future study.

The utilization of modelling techniques such as SEM (Structural Equation Modelling), powerful soft computing techniques, and other MCDM techniques such as DEMATEL are recommended in future studies to uncover the interdependencies between criteria and to provide different results. Due to the linkage of green university implementation and regional sustainable development (Wang et al. 2013), future in-depth research are encouraged to conduct on the relationships between sustainable development of the county and having more green universities in Malaysia. Moreover, making a comparison between the existing green building rating schemes and relevant credit criteria with GBI can be another direction for future investigations. This direction can deliver useful outcomes to develop the existing tools based on the local culture or even be a starting point to introduce new tools for green building assessment in developing nations.

Appendix 1: Green building index assessment criteria for NRNC (Greenbuildingindex 2009b)

See Tables 8, 9, 10, 11, 12 and 13.

Table 8 EE

Area of assessment	Item	Description	Max point
Design	EE1 MEEP	consider minimum EE to decrease consumption of energy in buildings and minimize the emission of CO ₂	1
	EE2 LIZ	To optimise energy savings, consider adaptable lighting systems	3
	EE3 ESM	Observe consumption of energy in basic building services and areas	5
	EE4 REE	Promote renewable energy usage	5
	EE5 AEEP	Surpass EE performance to minimize buildings' energy usage	15
Commissioning	EE6 ENC	Guarantee that building's energy-related systems are installed to attain appropriate commissioning	3
Verification and maintenance	EE7 POC	Provide post-occupancy commissioning in all tenancy zones	2
	EE8 EEV	Assess expected use of energy in main building services	2
	EE9 SUM	Make sure the energy-related systems in the building will remain working as intended	3
Total			35

Table 9 EQ

Area of assessment	Item	Description	Max point
Air quality	EQ1 MIAQP	Provide minimum IAQ performance to increase the occupants' well-being comfort and well-being	1
	EQ2 ETSC	Reduce indoor surfaces, occupants' exposure of building, and freshening air distribution structures to environmental tobacco smoke (ETS)	1
	EQ3 CDMC	Consider carbon dioxide's response monitoring levels	1
	EQ4 IAP	minimize detrimental effect on resident health from finishes which produce internal air chemicals	2
	EQ5 MOP	Provide mechanism that decreases the mould growth's risk and its related effects on resident health	1
Thermal comfort	EQ6 TCDCS	Offer the best level of thermal comfort structures by occupants or individuals to increase the building occupants' efficiency, comfort and well-being	2
	EQ7 ACE	Offer efficient clean air delivery by minimizing the indoor pollutants to boost a healthy indoor atmosphere	1
Lighting, visual and acoustic comfort	EQ8 DAY	Consider appropriate levels of building occupants' daylighting	2
	EQ9 DGC	Minimize discomfort of brightness from natural light	1
	EQ10 ELL	Control lighting of building office to not be over constructed	1
	EQ11 HFB	Improve amenity of workplace by preventing low-rate flicker	1
	EQ12 EXV	Decrease eyestrain for building occupants by provision of visual connection to the outdoor and permitting long-distance views	2
Verification	EQ13 INL	Preserve internal noise levels at a suitable degree	1
	EQ14 IAQ	Minimize IAQ challenges caused by construction activities to keep the building occupants' well-being and comfort	2
	EQ15 POCS	Support for the evaluation of building occupants' comfort	2
Total			21



Table 10 SM

Area of assessment	Item	Description	Max point
Site planning	SM1 SIS	Ignore designing hardscape, building, parking, or road zones on a site or part of a site, which cover any one of the predefined conditions	1
	SM2 BRR	Rehabilitate damaged sites by minimizing stress on undeveloped land	1
	SM3 DDCC	Connect expansion to urban region with present infrastructure, keep greenfield and maintain natural resources and habitat	2
	SM4 ENM	(A) protect current natural space and rebuild injured area to promote biodiversity and preserve habitat (B) Reduce Open Space by considering a high ratio of open space to improve footprint to encourage biodiversity	2
Construction management	SM5 ECAPC	Minimize construction activities' pollution by managing waterway sedimentation, soil erosion, and airborne dust production	1
	SM6 QLASSIC	Attain excellence of workmanship in construction activities	1
	SM7 WSA	Cut construction activities' pollution by managing pollution caused by rubbish and waste from labours	1
Transportation	SM8 PTA	Minimize land development effects and pollution from usage of automobile	1
	SM9 GVP	Promote green vehicles usage	1
	SM10 PAC	Prevent over-provision of car parking space	1
Design	SM11 SDQ	Restrict natural hydrology disruption by controlling storm water runoff, decreasing resistant cover, and enhancing on-site infiltration	1
	SM12 GR	Decrease heat island (thermal grade variation between undeveloped and developed zones) to cut effect on human, microclimate, and wildlife habitat	2
	SM13 BUM	Record characteristics and strategies of green building design for individual information and manage to maintain operation in the period of tenancy	1
Total			16

Table 11 MR

Area of assessment	Item	Description	Max point
Reused and recycled materials	MR1 MRS	Recycle building products and materials to diminish virgin materials' demand and decrease waste making	2
	MR2 RCM	Enhance request for building materials which include reused content materials in their production	2
Sustainable resources	MR3 REM	Utilize building products and materials which are produced within the region, thus encouraging the indigenous resources usage and cutting the environmental effects caused by transportation	1
Waste management	MR4 SUT	Promote forest management using environmentally friendly metrics	1
	MR5 SCR	Minimize generation of waste in the period of construction and even building usage	1
	MR6 CWM	Adopt a construction waste management strategy, as a minimum classifies the materials to be converted from removal regardless of if the materials will be placed on co-mingled or site	2
Green products	MR7 RCA	Utilize Refrigerants and Clean Agents which are environmentally friendly and increase commitment of Malaysia to the Montreal and Kyoto protocols	2
Total			11



Table 12 WE

Area of assessment	Item	Description	Max point
Water harvesting and recycling	WE1 RAH	Promote rainwater harvesting which will cause potable water consumption reduction	2
Increased efficiency	WE2 WAR	Promote water recycling which will cause potable water consumption reduction	2
	WE3 WEIL	Promote the development of system which is dependent on potable water supply usage from the local water agency	2
	WE4 WEF	Promote potable water consumption reduction by the utilization of effective methods	
	WE5 MLDS	Support the development of systems, which control and handle the consumption of water	2
Total			10

Table 13 IN

Area of assessment	Item	Description	Max point
	IN1 IDEDI	Offer design team and plan a strategy to be granted points for outstanding performance higher than the conditions considered by GBI rating scheme	6
	IN2 GBIAF	To promote the design combination needed for GBI rated buildings and to simplify the process of implementation and certification	1
Total			7

Appendix 2: Compassion matrices for criteria, sub-criteria and indicators

See Tables 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35 and 36.

Table 14 The results of main criteria pairwise comparison

	EE	EQ	SM	MR	WE	IN	Weight
EE	1	2.0	3.0	3.0	4.0	3.0	0.342
EQ		1	2.0	2.0	2.0	3.0	0.206
SM			1	3.0	2.0	4.0	0.180
MR				1	2.0	3.0	0.117
WE					1	4.0	0.101
IN						1	0.053
Incon: 0.07							

Table 15 The results of EE sub-criteria pairwise comparison

	Design	Commissioning	Verification and maintenance	Weight
Design	1	3	3	0.600
Commissioning		1	1	0.200
Verification and maintenance			1	0.200
Incon: 0.00				



Table 16 The results of design indicators pairwise comparison

	MEEP	LIZ	ESM	REE	AEEP	Weight
MEEP	1	0.5	1	0.5	0.2	0.092
LIZ		1	0.2	0.5	0.5	0.170
ESM			1	0.5	0.2	0.092
REE				1	0.5	0.226
AEEP					1	0.421
Incon: 0.02						

Table 17 The results of commissioning indicators pairwise comparison

	ENC	POC	Weight
ENC	1	3	0.75
POC		1	0.25
Incon: 0.00			

Table 18 The results of verification and maintenance indicators pairwise comparison

	EEV	SUM	Weight
EEV	1	0.25	0.20
SUM		1	0.80
Incon: 0.00			

Table 19 The results of EQ sub-criteria pairwise comparison

	Air quality	Thermal comfort	Lighting, visual and acoustic comfort	Verification	Weight
Air quality	1	2	0.33	2	0.236
Thermal Comfort		1	0.5	0.5	0.135
Lighting, visual and acoustic comfort			1	3	0.461
Verification				1	0.168
Incon: 0.08					

Table 20 The results of air quality indicators pairwise comparison

	MIAQP	ETSC	CDMC	IAP	MOP	Weight
MIAQP	1	1	1	0.5	1	0.167
ETSC		1	1	0.5	1	0.167
CDMC			1	0.5	1	0.167
IAP				1	1	0.333
MOP					1	0.167
Incon: 0.02						



Table 21 The results of thermal comfort indicators pairwise comparison

	TDCS	ACF	Weight
TDCS	1	3	0.75
ACF		1	0.25
Incon: 0.00			

Table 22 The results of lighting, visual & acoustic comfort indicators pairwise comparison

	DAY	DGC	ELL	HFB	EXV	INL	Weight
DAY	1	2.0	2.0	2.0	1.0	2.0	0.250
DGC		1	1.0	1.0	0.5	1.0	0.125
ELL			1	1.0	0.5	1.0	0.125
HFB				1	0.5	1.0	0.125
EXV					1	2.0	0.250
INL						1	0.125
Incon: 0.00							

Table 23 The results of verification indicators pairwise comparison

	IAQ	POSC	Weight
IAQ	1	1	0.50
POSC		1	0.50
Incon: 0.00			

Table 24 The results of SM sub-criteria pairwise comparison

	Site planning	Construction management	Transportation	Design	Weight
Site planning	1	2	2	2	0.395
Construction management		1	1	0.5	0.163
Transportation			1	0.5	0.163
Design				1	0.278
Incon: 0.02					

Table 25 The results of site planning indicators pairwise comparison

	SIS	BRR	DDCC	ENM	Weight
SIS	1	1	2	2	0.333
BRR		1	2	2	0.333
DDCC			1	1	0.167
ENM				1	0.167
Incon: 0.00					

Table 26 The results of construction management indicators pairwise comparison

	ECAPC	QLASSIC	WSA	Weight
ECAPC	1	1	1	0.333
QLASSIC		1	1	0.333
WSA			1	0.333
Incon: 0.00				



Table 27 The results of transportation indicators pairwise comparison

	PTA	GVP	PAC	Weight
PTA	1	1	1	0.333
GVP		1	1	0.333
PAC			1	0.333
Incon: 0.00				

Table 28 The results of design indicators pairwise comparison

	SDQ	GR	BUM	Weight
SDQ	1	0.5	1	0.25
GR		1	2	0.50
BUM			1	0.25
Incon: 0.00				

Table 29 The results of MR sub-criteria pairwise comparison

	Reused and recycled materials	Sustainable resources	Waste management	Green products	Weight
Reused and recycled materials	1	32	2	3	0.444
Sustainable resources		1	0.333	1	0.122
Waste management			1	3	0.312
Green products				1	0.122
Incon: 0.02					

Table 30 The results of reused and recycled materials indicators pairwise comparison

	MRS	RCM	Weight
MRS	1	1	0.50
RCM		1	0.50
Incon: 0.00			

Table 32 The results of Waste Management indicators pairwise comparison

	SCR	CWM	Weight
SCR	1	0.50	0.333
CWM		1	0.667
Incon: 0.00			

Table 31 The results of Sustainable Resources indicators pairwise comparison

	REM	SUT	Weight
REM	1	1	0.50
SUT		1	0.50
Incon: 0.00			

Table 33 The results of WE sub-criteria pairwise comparison

	Water harvesting and recycling	Increased efficiency	Weight
Water harvesting and recycling	1	0.25	0.20
Increased efficiency		1	0.80
Incon: 0.00			



Table 34 The results of water harvesting and recycling indicators pairwise comparison

	RAH	WAR	Weight
RAH	1	1	0.50
WAR		1	0.50
Incon: 0.00			

Table 35 The results of Increased Efficiency indicators pairwise comparison

	WEIL	WEF	MLDS	Weight
WEIL	1	1	1	0.333
WEF		1	1	0.333
MLDS			1	0.333
Incon: 0.00				

Table 36 The results of IN indicators pairwise comparison

	IDEDI	GBIAF	Weight
IDEDI	1	6	0.857
GBIAF		1	0.143
Incon: 0.00			

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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